SKILL - 10 React State Management using useState Object

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Skill Experiment Number: 10

Section Number: 75

App.jsx

```
import StudentManager from "./StudentManager";


function App() {
  return (
    <div>
      <StudentManager />
    </div>
  );
}


export default App;
```

Main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'


createRoot(document.getElementById('root')).render(
```

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```
  <StrictMode>
    <App />
  </StrictMode>,
)
```

StudentManager.css

```
.container{
  width:60%;
  margin:auto;
  text-align:center;
}

.form input{
  padding:8px;
  margin:5px;
}

button{
  padding:8px 12px;
  background:#007bff;
  color:white;
  border:none;
  margin:5px;
  cursor:pointer;
}
```

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```
.delete{
  background:red;
}

table{
  width:100%;
  border-collapse:collapse;
  margin-top:20px;
}

th,td{
  border:1px solid gray;
  padding:8px;
}
```

StudentManager.jsx

```
import { useState } from "react";
import "./StudentManager.css";

function StudentManager() {

  const initialStudents = [
    { id: 1, name: "Rahul", course: "CSE" },
    { id: 2, name: "Priya", course: "ECE" },
    { id: 3, name: "Arjun", course: "IT" },
    { id: 4, name: "Sneha", course: "AI" },
    { id: 5, name: "Kiran", course: "DS" }
```

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```
];

const [students, setStudents] = useState(initialStudents);

const [newStudent, setNewStudent] = useState({
  id: "",
  name: "",
  course: ""
});

const handleChange = (e) => {
  setNewStudent({
    ...newStudent,
    [e.target.name]: e.target.value
  });
};

const addStudent = () => {
  if (newStudent.id && newStudent.name && newStudent.course) {
    setStudents([...students, newStudent]);

    setNewStudent({
      id: "",
      name: "",
      course: ""
    });
```

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```
  }
};

const deleteStudent = (id) => {
  const updatedStudents = students.filter((student) => student.id !== id);
  setStudents(updatedStudents);
};

return (
  <div className="container">

    <h2>Student Manager</h2>

    <div className="form">
      <input
        type="text"
        name="id"
        placeholder="Enter ID"
        value={newStudent.id}
        onChange={handleChange}
      />

      <input
        type="text"
        name="name"
        placeholder="Enter Name"
```

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```
    value={newStudent.name}
    onChange={handleChange}
   />

   <input
    type="text"
    name="course"
    placeholder="Enter Course"
    value={newStudent.course}
    onChange={handleChange}
   />

   <button onClick={addStudent}>Add Student</button>
  </div>

  {students.length === 0 ? (
   <p>No students available</p>
  ) : (
   <table border="1" cellPadding="10">
    <thead>
     <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Course</th>
      <th>Action</th>
     </tr>
```

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```
        </thead>

        <tbody>
         {students.map((student) => (
           <tr key={student.id}>
             <td>{student.id}</td>
             <td>{student.name}</td>
             <td>{student.course}</td>
             <td>
               <button onClick={() => deleteStudent(student.id)}>
                 Delete
               </button>
             </td>
           </tr>
         ))}
        </tbody>

      </table>
    )}

  </div>
 );
}

export default StudentManager;
```

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```
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
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  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v4.0.3)

2026-03-10T07:44:15.447+05:30  INFO 17436 --- [studentmanager1] [           main] c.e.s.Studentmanager1Application
2026-03-10T07:44:15.466+05:30  INFO 17436 --- [studentmanager1] [           main] c.e.s.Studentmanager1Application
2026-03-10T07:44:17.263+05:30  INFO 17436 --- [studentmanager1] [           main] o.s.boot.tomcat.TomcatWebServer
2026-03-10T07:44:17.281+05:30  INFO 17436 --- [studentmanager1] [           main] o.apache.catalina.core.StandardService
2026-03-10T07:44:17.281+05:30  INFO 17436 --- [studentmanager1] [           main] o.apache.catalina.core.StandardEngine
2026-03-10T07:44:17.387+05:30  INFO 17436 --- [studentmanager1] [           main] b.w.c.s.WebApplicationContextInitializer
2026-03-10T07:44:18.116+05:30  INFO 17436 --- [studentmanager1] [           main] o.s.boot.tomcat.TomcatWebServer
2026-03-10T07:44:18.128+05:30  INFO 17436 --- [studentmanager1] [           main] c.e.s.Studentmanager1Application
```
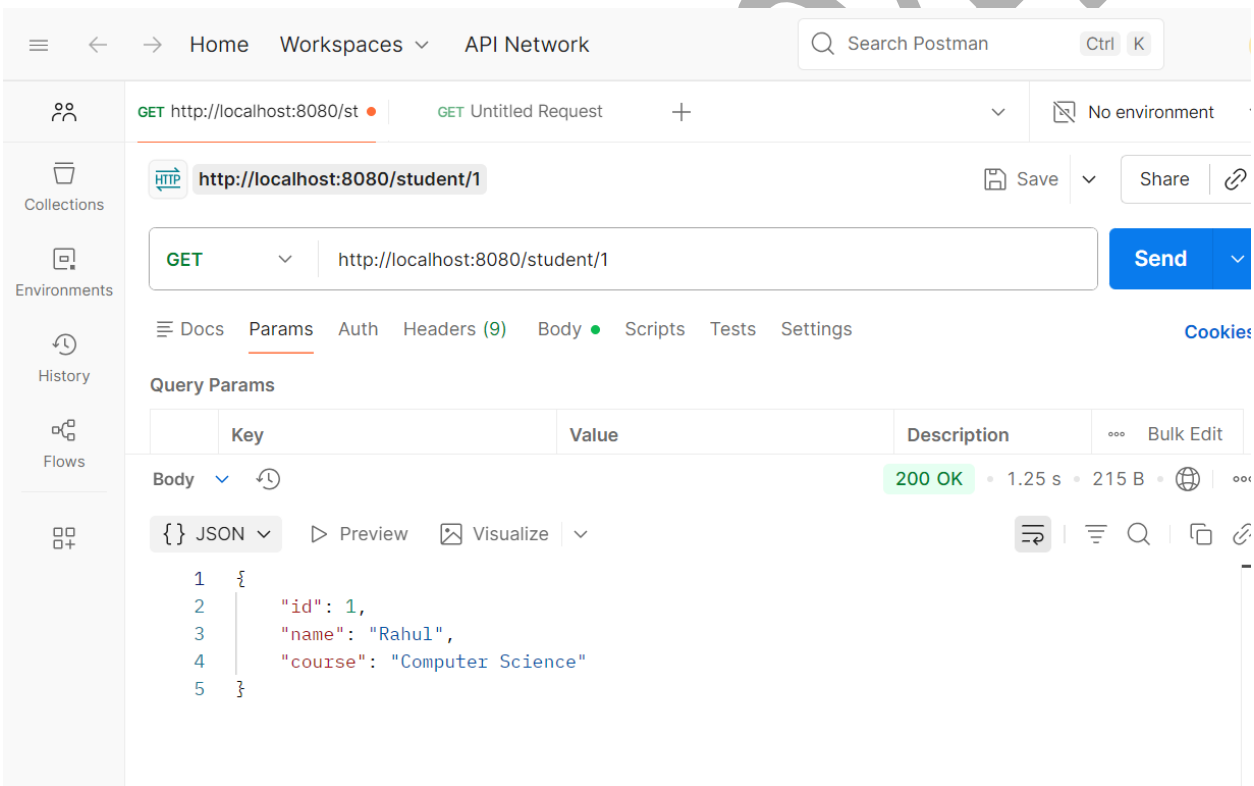

Output:



https://github.com/Gowtham-406/S-75_1041_SkillLabDocuments.git